



Network Security Jargon

Penetration Testing: Also called ethical hacking, it entails breaking into the security network of a system to assess its flaws

Vulnerability Analysis: Assess vulnerabilities of a system to security threats

Security Audit: Appraisal of the existing security framework of a company

Exploit Development: Exploiting existing vulnerabilities in a system in order to know all existing flaws in security

Teamwork At iViz

Team members keep up-to-date with the most recent exploits and vulnerabilities. They keep track of trends in intruder activities, analyse product vulnerabilities and malicious codes and record their research. De explains, "We develop customised security solutions based on our clients' needs. Our strength lies in penetration testing and security audit, and more so in exploitation development, vulnerability analysis and discovery." At the same time, De admits to having undertaken preliminary and at times not very challenging network security projects in order to sustain the company.

Managed Security Services

As the cornerstone of iViz' security solutions are Managed Security Services (MSS), which enable 24 x 7 monitoring and management of network security infrastructures. MSS comprise security audit, penetration testing, vulnerability analysis, security implementation and policy design. Through iViz' security information management architecture, critical security issues can be identified; clients can be alerted of impending danger; and appropriate and effective response measures can be prepared.

"MSS is a set of services that start with the detection of vulnerabilities, followed by their elimination and then continuous maintenance of the system. Although most firms spend huge amounts on security, they end up building a house with strong iron doors and walls made of straw. MSS can equip them with robust security at a lesser cost," says Barai.

Looking Beyond Firewalls

Popular belief says that a firewall and an Intrusion Detection System (IDS) can keep an organisation safe. Not so, claims Barai. "In all the sensational hacking cases we have seen, the affected parties had firewalls and IDSes. We look beyond firewalls and IDSes to find vulnerabilities in the human and computer interface—one of the weakest links in the chain, which determines the effectiveness of security. We help organisations become secure from all known vulnerabilities."

Penetration Testing

A part of MSS, penetration testing is popularly known as ethical hacking. An established technique for the live testing of network security, penetration testing attempts to compromise a system, as a potential attacker would, to see what kind of information is actually divulged, thus determining the vulnerability of a system. It enables companies to receive a real-life vulnerability test of their security status.

The iViz team has exposed several vulnerabilities and exploits that were hitherto unknown. "A valuable testing, auditing and evaluation tool,



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Chief Technology Officer,
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penetration testing can be conducted remotely over the Internet and does not require physical access to the premises or any privileged information such as usernames or passwords. It involves detection of vulnerabilities, their removal, and then, continuous maintenance of the system," says De. For a network with three to four servers and 100-odd users, the iViz team can complete penetration testing in three days.

Bramhastra To The Rescue

iViz has built a penetration testing tool called *Bramhastra*. "Bramhastra simulates an attack that can be launched by the worst possible malicious user. It tests the security of an organisation by actually breaking into the system, gives a report and suggests measures to correct the flaws and finally, it clears up all the traces of the test," reveals Barai and claims that *Bramhastra* can break into the toughest of networks.

Bramhastra is supposed to be able to break into the toughest of networks. New exploits are constantly added to keep the *Bramhastra* updated. iViz claims to be the sole Indian company to have developed such a tool.

Blocking With Spam Walls

iViz also has a patented algorithm to detect spam that passes through conventional filters and is provably more efficient than other solutions such as 'Bright mail Anti Spam Enterprise Edition', 'MacAfee Spam Killer' and 'Postini Perimeter Manager'. 'Spam Wall' is a unique anti-spam solution from iViz that can be used for individual and server installations. It is a hardware-based filter that sits between the mail server and the Internet. Spam Wall is platform-independent, and one just has to plug-and-play. Once installed, it takes a week to resolve the spamming problem.

A major issue with existing anti-spam solutions is the large number of false positives returned (valid e-mails being stopped as spam). "With 'Spam Wall' we already ensure extremely low false positives. Hopefully, we will be able to guarantee zero false positives in a few months," assures Barai.

The Road Ahead

iViz conducts its core research work and product development at IIT-K, and has a marketing and corporate presence in Kolkata and Texas, USA. Currently, iViz is looking for venture funding for further expansion. It has job openings for professionals with a deep knowledge of security fields and a zest for innovation.

iViz wants to make computing secure at an affordable price. Barai says, "We plan to market our products shortly. In fact, the beta versions are ready. However, we are not sure about marketing the *Bramhastra*, as it could be very dangerous if it falls into the wrong hands." The team will soon provide training on computer security.

"This is just the beginning and the best is yet to come," reflects De. With its grand vision of cordoning off the increasingly susceptible virtual space from anti-social elements, the student-turned-entrepreneurs of iViz Techno Solutions will have to keep up the good work. ■

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